

Overview

1 Initial value problems for pdes

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- The FTCS scheme
- Improved discretisation schemes
- Implicit schemes
- The Crank–Nicolson scheme
- Schrödinger equation

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3 Summary

Boundary and initial value problems

We looked at boundary value problems, which may be solved in a similar way to boundary-value odes.

Now: **initial value problems**.

Start with diffusion equation in 1+1 dimension:

Diffusion equation

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$$

Identical to **heat equation**.

Similar to **Schroedinger equation** in 1+1 dimension.

Initial value problems for pdes

Diffusion equation in 1+1 dimension:

$$\frac{\partial u}{\partial t} = D \frac{\partial^2 u}{\partial x^2}$$

Discretise space & time.

Using grid spacings $\Delta x = a$ in space, Δt in time, we write

$$u(x_i, t_n) = u(x_0 + ia, t_0 + n\Delta t) \equiv u_i^{(n)}.$$

Take **forward derivative** in time, and **symmetric second derivative** in space:

Forward **T**ime, **C**entred **S**pace

$$\begin{aligned} \frac{\partial u}{\partial t} &= \frac{u_j^{(n+1)} - u_j^{(n)}}{\Delta t} = D \frac{\partial^2 u}{\partial x^2} = \frac{D}{a^2} \left(u_{j-1}^{(n)} - 2u_j^{(n)} + u_{j+1}^{(n)} \right) \\ \implies u_j^{(n+1)} &= \left(1 - \frac{2\Delta t}{a^2} \right) u_j^{(n)} + \frac{\Delta t}{a^2} \left(u_{j+1}^{(n)} + u_{j-1}^{(n)} \right) \end{aligned} \quad (1)$$

It is straightforward to iterate this forward in time from $n = 0$.

Stability analysis

von Neumann analysis

Fourier transform in space: $u(x) = \sum_k e^{ikx} u(k)$

Each $u(k)$ evolves independently in time

(at least for linear problems with constant coeffs)

This gives the **eigenmode evolution**

$$u_k^{(n+1)} = \xi_k u_k^{(n)} \quad \implies \quad u_j^{(n)} = u_0^{(0)} \xi_k^n e^{ikja} \quad (2)$$

To find **amplification factor** ξ_k , substitute (2) into finite difference equation

$$\xi_k \begin{cases} > 1 & \text{exponential growth, instability} \\ < 1 & \text{exponential damping, stability} \\ = 1 & \text{more detailed analysis needed} \end{cases}$$

von Neumann stability: $|\xi_k| \leq 1 \forall k$

Stability for FTCS

Inserting the eigenmode evolution (2) into the FTCS equation (1) gives

$$\frac{\xi_k^{n+1} e^{ijka} - \xi_k^n e^{ijka}}{\Delta t} = \frac{D}{a^2} \xi_k^n \left(e^{ik(j-1)a} - 2e^{ikja} + e^{ik(j+1)a} \right)$$

$$\xi_k = 1 + \frac{D\Delta t}{a^2} \left(e^{-ika} - 2 + e^{ika} \right) = 1 - \frac{4D\Delta t}{a^2} \sin^2 \frac{ka}{2}$$

This is always $\leq 1 \quad \Rightarrow \quad$ stability requires $\xi_k \geq -1$

$$\Rightarrow \frac{4D\Delta t}{a^2} \sin^2 \frac{ka}{2} \leq 2.$$

'Worst case': $\sin^2(ka/2) = 1$

$$\text{Stability condition: } \frac{\Delta t}{a^2} \leq \frac{1}{2D}$$

An example

The python file `ftcs_driver.py` solves the diffusion equation with the initial distribution

$$u_0(x) = u(x, t_0) = e^{-x^2/4Dt_0}, \quad -5 \leq x \leq 5, t_0 = 0.1,$$

and boundary conditions

$$u(\pm x_0, t) = \sqrt{\frac{t_0}{t}} e^{-x_0^2/4Dt}, \quad x_0 = 5.$$

The grid spacing in the x direction has been set to $a = 0.05$, and the diffusion constant $D = 1$.

`ftcs_driver(dt, t)` plots the solution for time step dt at time(s) t .

Run this with $dt=0.0012$ and see what you get.

Then run with $dt=0.0013$ and see what happens.

FTCS in 2+1 dimension

Our Ansatz is now

$$u_{jl}^{(n)} = u_0 \xi_k^n e^{ik_x j \Delta x} e^{ik_y l \Delta y} \quad (3)$$

For $\Delta x = \Delta y = a$ the FTCS scheme is

$$\frac{u_{jl}^{(n+1)} - u_{jl}^{(n)}}{\Delta t} = \frac{D}{a^2} \left(u_{j-1,l}^{(n)} + u_{j,l-1}^{(n)} + u_{j+1,l}^{(n)} + u_{j,l+1}^{(n)} - 4u_{jl}^{(n)} \right)$$

Inserting (3) gives

$$\begin{aligned} \xi_k &= 1 + \frac{D\Delta t}{a^2} \left(e^{-ik_x a} + e^{-ik_y a} + e^{ik_x a} + e^{ik_y a} - 4 \right) \\ &= 1 - \frac{4D\Delta t}{a^2} \left(\sin^2 \frac{k_x a}{2} + \sin^2 \frac{k_y a}{2} \right) \end{aligned}$$

$$|\xi_k| \leq 1 \forall k \implies \frac{\Delta t}{a^2} \leq \frac{1}{4D}$$

Improving stability (want to use larger Δt)

Stability condition is quite severe



- We want to model features at large scales $\lambda \gg a$
Typical diffusion time is $\tau \sim \lambda^2/D$
→ need $n = \frac{\tau}{\Delta t} \sim \frac{\lambda^2}{a^2}$ time steps
- We want to improve accuracy by reducing a
But if $a \rightarrow a/2$ then $\Delta t \rightarrow \Delta t/4$
→ 8 times as much cpu time!

Can we improve on this?

Second order time derivative?

FTCS is first-order accurate in time, second order in space

What about using second-order differencing in time?

Centred Time Centred Space

$$\frac{u_j^{(n+1)} - u_j^{(n-1)}}{2\Delta t} = \frac{D}{a^2} \left(u_{j-1}^{(n)} - 2u_j^{(n)} + u_{j+1}^{(n)} \right)$$

von Neumann

$$\xi_k - \frac{1}{\xi_k} = -\frac{8D\Delta t}{a^2} \sin^2 \frac{ka}{2}$$
$$\Rightarrow \xi_k = -\frac{4D\Delta t}{a^2} \sin^2 \frac{ka}{2} \pm \sqrt{1 + \left(\frac{4D\Delta t}{a^2} \sin^2 \frac{ka}{2} \right)^2}$$

The $(-)$ mode is unstable for **all** k and Δt !

CTCS is unconditionally unstable

Implicit schemes: BTCS

Explicit scheme: $\frac{\partial^2 u}{\partial x^2}$ evaluated at t

Implicit scheme: evaluate at $t + \Delta t$

Backward Time, Centred Space

$$\frac{u_j^{(n+1)} - u_j^{(n)}}{\Delta t} = \frac{u_{j-1}^{(n+1)} - 2u_j^{(n+1)} + u_{j+1}^{(n+1)}}{a^2}$$
$$\implies \left(1 + \frac{2\Delta t}{a^2}\right) u_j^{(n+1)} - \frac{\Delta t}{a^2} \left(u_{j-1}^{(n+1)} + u_{j+1}^{(n+1)}\right) = u_j^{(n)}$$

We get a sparse matrix equation for $u^{(n+1)}$.

von Neumann analysis

$$\frac{\xi - 1}{\Delta t} = \frac{\xi}{a^2} \left(e^{ika} - 2 + e^{-ika}\right) \implies \xi = \frac{1}{1 + \frac{4\Delta t}{a^2} \sin^2 \frac{ka}{2}}$$

$\xi < 1$ for **all** $k, \Delta t$: BTCS is **unconditionally stable**

Crank–Nicolson

BTCS is stable, but only first-order accurate in time.

How can we get second-order accuracy?

Average FTCS and BTCS!

(the same as taking a centred time derivative around $t + \Delta t/2$)

$$\frac{u_j^{(n+1)} - u_j^{(n)}}{\Delta t} = \frac{1}{2a^2} \left[\underbrace{u_{j+1}^{(n+1)} - 2u_j^{(n+1)} + u_{j-1}^{(n+1)}}_{BTCS} + \underbrace{u_{j+1}^{(n)} - 2u_j^{(n)} + u_{j-1}^{(n)}}_{FTCS} \right]$$

Crank–Nicolson

$$\left(1 + \frac{\Delta t}{a^2}\right) u_j^{(n+1)} - \frac{\Delta t}{2a^2} (u_{j-1}^{(n+1)} + u_{j+1}^{(n+1)}) = \left(1 - \frac{\Delta t}{a^2}\right) u_j^{(n)} + \frac{\Delta t}{2a^2} (u_{j-1}^{(n)} + u_{j+1}^{(n)})$$

Stability of Crank-Nicolson

von Neumann analysis

$$\xi \left(1 - \frac{-4\Delta t}{2a^2} \sin^2 \frac{ka}{2} \right) = 1 + \frac{-4\Delta t}{2a^2} \sin^2 \frac{ka}{2}$$

$$\Rightarrow \boxed{\xi = \frac{1 - \frac{2\Delta t}{a^2} \sin^2 \frac{ka}{2}}{1 + \frac{2\Delta t}{a^2} \sin^2 \frac{ka}{2}}} = \frac{1 - b^2}{1 + b^2}$$

The modulus of the numerator is always smaller than the denominator

Crank-Nicolson is unconditionally stable

Schrödinger equation

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi + V(\vec{x})\Psi$$

This can be discretised in the same way as diffusion equation

Important constraint

Time evolution is **unitary** $\iff$ Total probability is conserved

Formally: $\Psi(t) = e^{iH(t-t_0)/\hbar} \Psi(t_0)$

FTCS for Schrödinger equation

$$\Psi^{(n+1)} = \Psi^{(n)} - \frac{i\Delta t}{\hbar} H \Psi^{(n)} = \left(1 - \frac{i\Delta t}{\hbar} H\right) \Psi^{(n)}$$

This is unconditionally unstable, and not unitary!

BTCS is **stable**, but still not unitary.

A unitary evolution operator

Use the **Cayley form** for $\exp(iH)$,

$$e^{iH\Delta t} \approx \frac{1 - \frac{1}{2}iH\Delta t}{1 + \frac{1}{2}iH\Delta t}$$

This gives us

$$\left(1 + \frac{1}{2}iH\Delta t\right)\psi^{(n+1)} = \left(1 - \frac{1}{2}iH\Delta t\right)\psi^{(n)}$$

This is Crank–Nicolson again!

Stable, second-order accurate and **unitary**.

Wave equation

Examples too numerous to list. . .

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

has formal solution

$$u(x, t) = F(x - ct) + G(x + ct), \quad \text{any } F, G$$

The equation can be discretised directly (using **staggered leapfrog** scheme) or transformed to two coupled first-order equations,

$$\begin{aligned} \frac{\partial v}{\partial t} + c \frac{\partial w}{\partial x} &= 0, & \frac{\partial w}{\partial t} + c \frac{\partial v}{\partial x} &= 0 \\ \text{with } v(x, t) &= \frac{\partial u}{\partial t}, & w(x, t) &= -c \frac{\partial u}{\partial x} \end{aligned}$$

Courant–Friedrichs–Lewy condition

Any explicit discretisation scheme for **wave** or **advection** equation requires

$$\Delta t < c \Delta x$$

Summary

- von Neumann stability criterion for time evolution equations:
 - ① Fourier transform in space
 - ② Find amplification factor ξ_k for each mode k
 - ③ Stability $\iff \xi_k \leq 1$
- Forward Time, Centred Space:
Explicit, stable for $\Delta t \leq a^2/2dD$ (1+d dim);
- Backward Time, Centred Space:
Implicit, unconditionally stable
- Both FTCS and BTCS are first order in time
- Crank–Nicolson: Average FTCS and BTCS.
Second order in time, unconditionally stable